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Job Data Pipeline



Presented by: Team C

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Client & Project Context



Client

- Focus on technology jobs in Saudi Arabia.
- Organizations need better visibility into technology hiring trends.
- Job information is available across different platforms.
- Understanding market demand can support better decision-making.

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Client & Project Context



The project focuses on:

- Collecting job postings from multiple sources
- Combining the data into one dataset
- Cleaning and standardizing job information
- Creating a centralized, analysis-ready dataset

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Pain Points

- **Fragmented Data**

Job postings are distributed across multiple sources.

- **Inconsistent Formats**

Different sources use different structures and formats.

- **Inconsistent Information**

Job titles, skills, locations, and company names can vary.

- **Data Quality Issues**

Missing, duplicate, or outdated job postings.

- **Arabic & English Content**

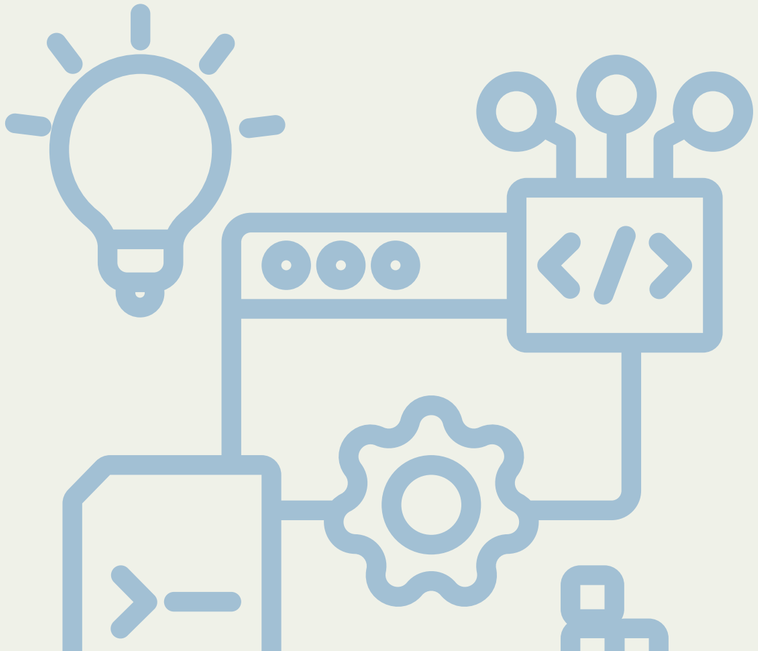
Job information may appear in different languages.



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Proposed Solution



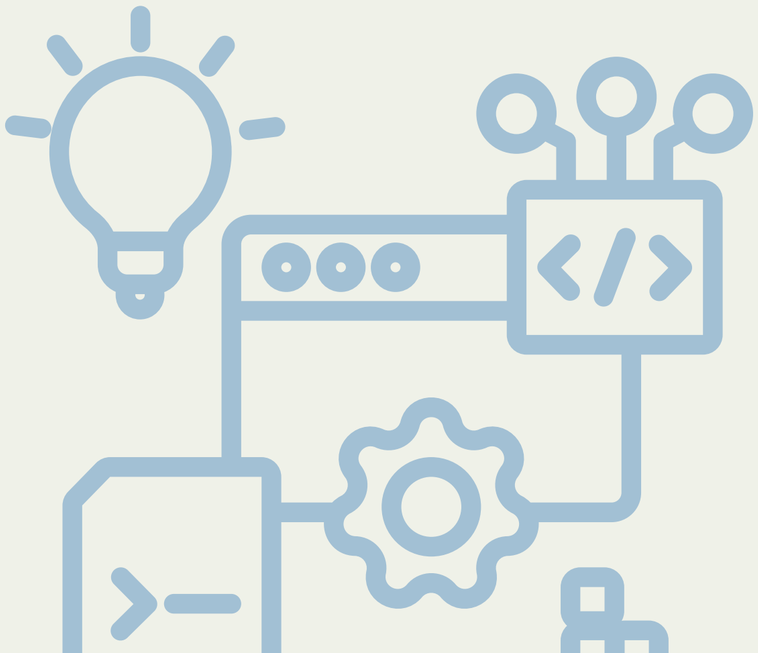
1 What We Are Building

- One automated, repeatable pipeline for Saudi tech job data
- Collects postings from multiple sources: Jooble, JSearch, FreeHire, Tapneo, Tanqeeb
- Built to handle real-world issues: inconsistent formats, missing data, duplicate posts
- Turns scattered postings into one standardized structure

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Proposed Solution



2 What It Delivers

- Organizes data through Bronze → Silver → Gold layers
- Cleans, deduplicates, and validates every record before it's final
- Runs automatically on a schedule — no manual work needed
- Delivers one reliable, structured, analysis-ready dataset
- A ready foundation for future dashboards and market analysis

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Solution Architecture

Extract
Python ,Requests

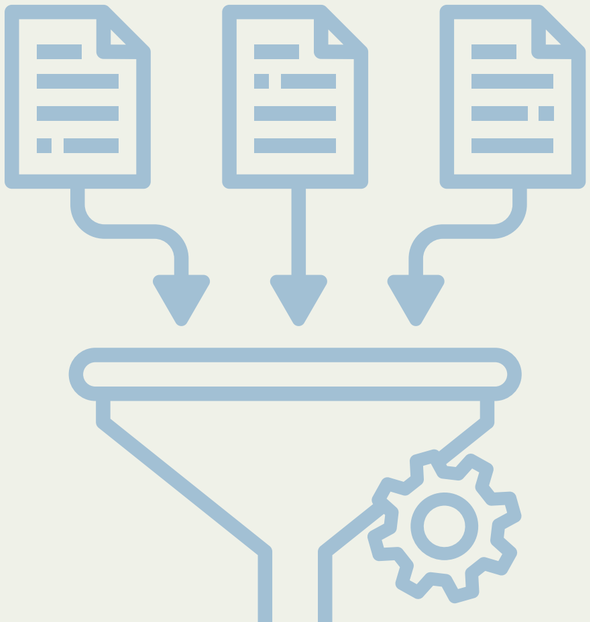
Bronze [Raw]
ADLS Gen2 Storage

Clean and Validated
Pandas,Rapidfuzz

Silver[Clean]
Clean and Standardized

Aggregation
Combine All Sources

Gold [Ready]
Ready for BI, analysis

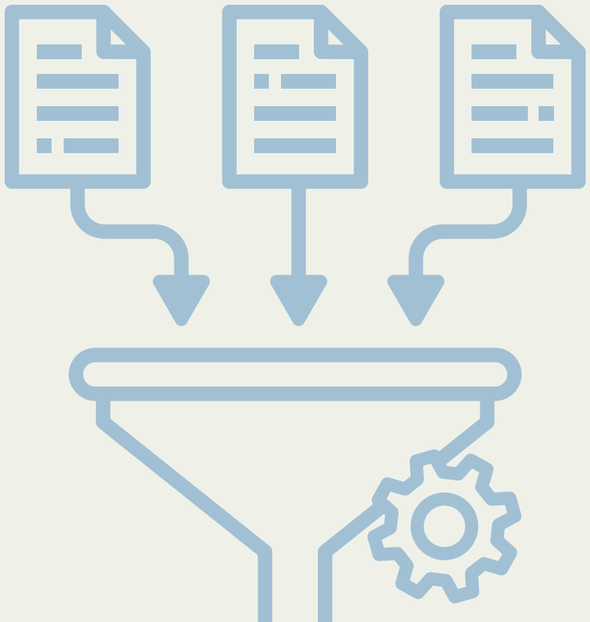


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Sources

Source	Access	Format	Source link
Jooble	Official API (POST request with a custom key)	JSON	https://jooble.org/api/
JSearch	API via RapidAPI (GET request)	CSV / JSON	rapidapi.com
FreeHire	API Request via Browser	JSON	https://freehire.me/
Tapneo	Static Dataset Download + Pandas Read	CSV/JSON file	tapneo-data.com
Tanqeep	Web Scraping (Via ScrapeOps Proxy API)	JSON (Semi-Structured Data)	https://saudi.tanqeeb.com/ar



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Implementation Plan

1 Multi-Source Ingestion & Raw Storage

- Collect job postings across target sources
- Ingest and store immutable raw files directly in Azure Data Lake Storage (ADLS) Gen2 to ensure auditability and future reprocessing



2 Cleaning & Schema Standardization

- Read raw files using Python & Pandas to map varying source attributes into a unified Common Schema
- Sanitize text fields, handle missing values, and normalize Saudi city locations

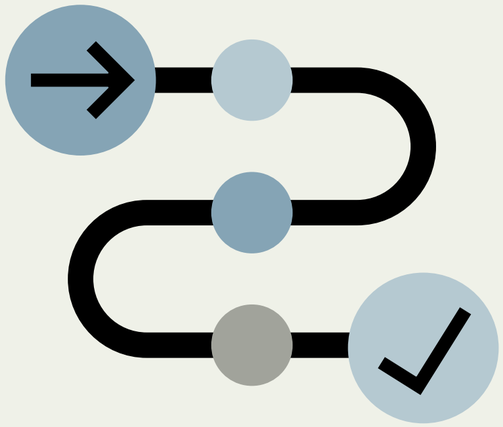
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Implementation Plan

3 Data Quality & Validation Checks

- Validate data types, detect corrupted entries, and ensure values conform to expected Saudi market criteria (e.g., valid cities, realistic ranges) before final storage



4 Curated Storage & Preparation

- Store cleaned and standardized records in ADLS Gen2 as separate files for each source within the Silver Layer
- Aggregate these records into the Gold Layer as a unified dataset optimized for Power BI dashboards, city-level querying, and downstream market analysis

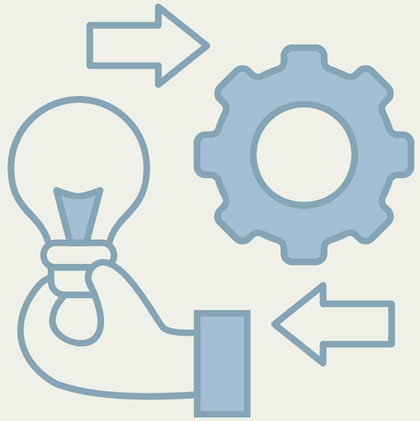
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Implementation Plan

5 Automated Scheduling & Pipeline Orchestration

- Configure Azure Functions (Timer Trigger) for automated, hands-off pipeline execution.
- Enable periodic ingestion and processing to keep job-market datasets continuously up to date.
- Ensure seamless end-to-end execution from raw source ingestion to final storage



6 Downstream Extensions (Time-Permitting)

- Build business-level aggregations and connect an interactive dashboard (e.g., Streamlit or Power BI) to visualize top skills, demand trends

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Timeline



WEEK THREE (2 - 5 September) :

- Build & manually test extraction scripts (Python + requests) for Jooble, Tanqeeb, RapidAPI
- Automate with Azure Functions — runs every 6 hours
- Output: 3 raw files → Bronze layer

WEEK FOUR (6 - 12 September) :

- Clean each Bronze file individually using Pandas
- Test manually, then automate with Azure Functions
- Output: each cleaned file → Silver layer separately

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Timeline



WEEK FIVE (13 - 19 September) :

- Merge the three Silver files (Pandas + RapidFuzz)
- Remove duplicates, run quality checks
- Automate, triggered after Stage 2
- Output: one final file → Gold layer

WEEK SIX (20 - 26 September) :

- End-to-end test of the full 3-function pipeline
- Complete documentation
- Final presentation ready before Oct 1

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Thank You

We welcome your questions and feedback

